



Linear frequency swept laser source with high swept slope based on digital optical phase-locked loop

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ABSTRACT

We present a method for generating linearly frequency swept optical waves using a digital optical phase-locked loop. The slope and shape of the optical frequency swept is locked to a constant reference frequency, which is controlled by a program. The method enables precise control of lasers with nonlinear tuning characteristics, and achieves a high linearity of frequency swept optical wave, thus providing a stable and highly coherent frequency swept optical source for laser ranging, velocimetry and 3D imaging. Experimentally, we achieve a frequency swept bandwidth of 31.1 GHz with a frequency swept slope of 518 THz/s using a distributed feedback laser diode (DFB-LD). In addition, we analyzed the linearity of the frequency-swept optical wave after phase-locking, and the root mean square (RMS) value of the frequency swept nonlinearity was below 1.4 MHz.

1. Introduction

Linearly frequency-swept laser sources have a wide range of applications in optical coherence tomography (OCT) [1–3] and optical frequency-domain reflectometry (OFDR) [4–6], especially in frequency modulated continuous wave (FMCW) light detection and ranging (LiDAR) technology [7–9]. FMCW LiDAR utilizes the coherent detection principle to extract frequency information [10,11], thus avoiding interference from sunlight and other LiDAR emitters [12], and it can also acquire both position and velocity information of the target in a single measurement [13]. The linear frequency swept laser source is the key component of FMCW LiDAR, and the maximum distance that can be measured by LiDAR is determined by the coherence of the laser source [9], while the spatial resolution depends on the frequency swept bandwidth of the laser [9] source. Usually, the laser that allows a large frequency swept bandwidth has worse phase noise characteristics, while a laser source with excellent coherence usually does not have good frequency tuning characteristics. In addition, the nonlinearity of the laser in the frequency swept process also limits the measurement distance and spatial resolution. Therefore, a frequency swept laser source with excellent coherence, high linearity, and a frequency swept bandwidth that meets the application requirements becomes the key to FMCW LiDAR research.

Generally, the approaches to laser implementation of linear frequency swept can be classified into two categories: (i) externally modulated approaches [14], which are achieved by connecting an optical

frequency modulator, such as an acoustic-optical modulator, and (ii) internally modulated approaches [15], which are achieved by controlling the injection current and temperature of the tunable laser. For implementing a high linearity frequency swept laser source, the most effective method is to employ an externally modulated semiconductor laser (SCL) approach, but this method requires an expensive broadband arbitrary waveform generator (AWG) to generate a linear frequency swept electrical signal, thus limiting the application of this method [14]. For the internal modulation method, most semiconductor lasers do not have a linear relationship between optical frequency and tuning signal [15,16], especially the tuning signal with high repetition rate will cause the laser to exhibit greater nonlinear characteristics, and various linearization methods are needed to achieve linear frequency swept. However, this modulation method does not require expensive AWGs and is easy to integrate, so it has obtained wide attention.

The internal modulation method to achieve frequency swept linearization of semiconductor lasers, the simplest way is to solve the nonlinear relationship between the modulation current and the laser output frequency. There are two mainstream solutions, one is to employ iterative pre-distortion algorithm [17,18]; the other is to employ optical phase-locked loop (OPLL) for real-time active frequency swept linearization [19–21]. The iterative pre-distortion algorithm can effectively improve the frequency swept nonlinearity of the semiconductor laser, but it is unable to suppress the broadband random frequency noise [22], and thus cannot improve the coherence characteristics of

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the laser. The OPLL can effectively suppress the random frequency noise, but it is limited by the phase-locked bandwidth, which makes it difficult to track a large range of nonlinear frequency errors.

To address the problems described above, we propose a digital phase-locked-loop implementation of frequency swept linearization, which incorporates a pre-distortion algorithm, which can be used to improve the linearity and coherence of a frequency swept laser source. The iterative pre-distortion algorithm is used for the generated pre-distortion drive current, which can be used to correct the frequency swept nonlinearity of the laser over a wide range, and the phase-locked loop is used to suppress the remaining frequency swept nonlinearity. This linear frequency swept approach has no specific requirements on the laser. Experimentally, we achieve a frequency swept bandwidth of 31.1 GHz in 120 μ s using a DFB-LD, and the RMS value of the frequency swept nonlinearity after phase-locking is less than 1.4 MHz. Since the dynamic linewidth is beyond the range of our current research, the change in the dynamic linewidth of the laser before and after calibration is not given in this paper. Using this method, a high-performance long-term stable FMCW LiDAR system can be obtained. And it does not require expensive linear frequency swept lasers, complicated linearization setups, and tedious data post-processing. According to theoretical calculations, the maximum detection distance that can be achieved by this LIDAR system is 272 m.

2. Principle of digital optical phase locked technology

Fig. 1 shows a digital optical phase-locked system that implements a linear frequency swept. The system incorporates an iterative pre-distortion algorithm. Before the system is closed loop, the drive current waveform obtained by the iterative pre-distortion algorithm needs to be first pre-input into the program control circuit board (PCCB). After the system is closed, the DFB-LD outputs a periodic frequency swept optical wave, and its time-frequency curve is shown in Fig. 1(a). After passing through the isolator, the frequency swept optical wave is divided into two light beams by a 90:10 coupler. 90% of these optical waves are output directly, and the other 10% are sent to an unbalanced fiber Mach-Zehnder interferometer (MZI) to produce autocorrelation external differential interference. The time-frequency curves of the two-frequency swept optical waves with delay τ are shown in Fig. 1(b). The photodetector (PD) converts the two coherently superimposed optical waves into electrical signals, which enter the PCCB after the analog-to-digital converter (ADC). The PCCB mainly consists of a field programmable gate array (FPGA) and a microcontroller unit (MCU). The signal is digitally mixed and filtered in the FPGA, then digitally operated in the MCU and generates a phase error signal, which is superimposed with a pre-distortion waveform preset in the FPGA and then acted on the laser driver after passing through a digital-to-analog converter (DAC). In the closed-loop locked state, the phase error tends to zero, which means that the DFB-LD output is a stable linear swept optical waveform.

In the above feedback system, the frequency of the photocurrent generated by the PD is proportional to the frequency swept slope of the optical wave, and the frequency of the photocurrent should be locked to the frequency of the reference signal output by the PCCB. The frequency swept slope of the DFB-LD output optical wave can be changed by changing the frequency of the reference signal in the locked state. The pre-distortion signal, reference signal, and system control signal are all implemented by PCCB to ensure the stable operation of the feedback system.

Suppose that the electric field of the frequency swept optical wave at the input (1) of MZI in Fig. 1 is represented as

$$E(t) = a \cos [\omega_{SFL}(t) \cdot t + \theta] \quad (1)$$

where a denotes the field amplitude, $\omega_{SFL}(t)$ is the instantaneous frequency of the DFB-LD output optical wave, and θ is the initial phase of the optical wave. The phase in Eq. (1) is denoted as $\phi_{SFL}(t)$.

$$\phi_{SFL}(t) = \omega_{SFL}(t) \cdot t + \theta \quad (2)$$

The linear frequency swept requires that the instantaneous frequency $\omega_{SFL}(t)$ of the DFB-LD output optical wave varies linearly with time, i.e.

$$\omega_{SFL}(t) = \omega_0 + 2\pi\xi_d \cdot t \quad (3)$$

where ω_0 is the optical carrier frequency, and ξ_d is the linear frequency swept slope, both of which are constants.

At the unbalanced MZI outgoing end, the two optical fields are represented by the Eq. (4).

$$E_1(t) = \frac{\sqrt{2}}{2} E(t - \tau) = \frac{\sqrt{2}}{2} a \cos [\omega_{SFL}(t - \tau) \cdot (t - \tau) + \theta] \quad (4)$$

$$E_2(t) = \frac{\sqrt{2}}{2} E(t) = \frac{\sqrt{2}}{2} a \cos [\omega_{SFL}(t) \cdot t + \theta]$$

where τ is the fiber delay time of the unbalanced MZI.

$$I(t) = \left\{ \frac{\sqrt{2}a}{2} \cos [\omega_{SFL}(t - \tau) \cdot (t - \tau) + \theta] + \frac{\sqrt{2}a}{2} \cos [\omega_{SFL}(t) \cdot t + \theta] \right\}^2 \quad (5)$$

The photocurrent signal output from PD is proportional to the optical intensity, and the high frequency component in Eq. (5) is beyond the response range of the PD, and when the photocurrent signal passes through the band-pass filter (BPF), the expression of the photocurrent signal output from PD contains only the differential frequency component, as shown in Eq. (6).

$$\begin{aligned} i_{beat}(t) &\propto \frac{Ra^2}{2} \cos[\omega_{SFL}(t) \cdot t - \omega_{SFL}(t - \tau) \cdot (t - \tau)] \\ &\propto \frac{Ra^2}{2} \cos[\phi_{SFL}(t) - \phi_{SFL}(t - \tau)] \end{aligned} \quad (6)$$

where R is the responsiveness of the PD.

After the PD output beat signal is passed through ADC, the analog signal is converted into digital signal and input to PCCB, which is mainly used as digital average discriminator in the system, and the principle of digital average discriminator is shown in Fig. 2.

The digital controlled oscillator (DCO) generates the in-phase and quadrature signals I and Q , respectively. The beat signals of PD are multiplied with I and Q through a digital multiplier (MUL), and then passed through a low-pass filter (LPF) respectively as

$$\begin{aligned} i_{beat}(t) \cdot I &= \frac{Ra^2}{2} \cos[\phi_{SFL}(t) - \phi_{SFL}(t - \tau) - \omega_R t] = \cos \theta_e \\ i_{beat}(t) \cdot Q &= \frac{Ra^2}{2} \sin[\phi_{SFL}(t) - \phi_{SFL}(t - \tau) - \omega_R t] = \sin \theta_e \end{aligned} \quad (7)$$

where ω_R is the oscillation frequency of the DCO reference signal, and θ_e is the phase error signal.

Both signals of Eq. (7) complete the division operation in the MCU.

$$\tan \theta_e = \frac{\sin \theta_e}{\cos \theta_e} \quad (8)$$

The phase error is

$$K \theta_e = K \cdot \tan^{-1} \left(\frac{\sin \theta_e}{\cos \theta_e} \right) \quad (9)$$

The phase error signal is used as the loop feedback quantity, where K is the gain coefficient, which is used to tune the magnitude of the loop feedback quantity. In the experiment, we will determine the range of the gain coefficient by estimation, then ensure the other parameters are fixed, adjust the value of the gain coefficient K continuously, and observe the time-frequency curve of the beat signal to determine the optimal value of the gain coefficient. The phase error signal is superimposed with the pre-distortion signal preset in the PCCB through the adder, and then act together to the driver of the DFB-LD to make the system form a closed loop, when the loop is locked $\omega_{PD} = \omega_R = \xi_d \tau$, θ_e converges to zero, ω_{PD} is the frequency of the beat signal.

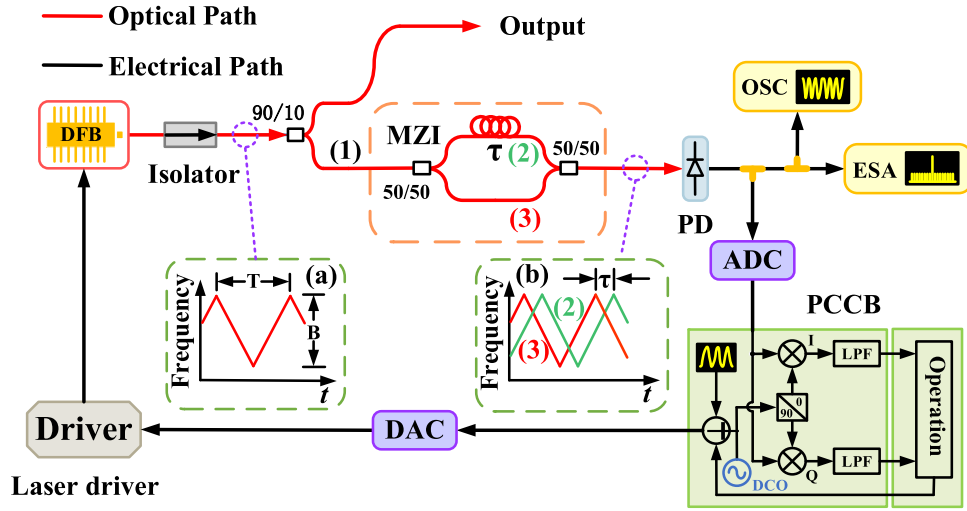


Fig. 1. Digital optical phase-locked system. DFB: distributed feedback laser; MZI: Mach-Zehnder interferometer; PD: photodetector; OSC: oscilloscope; ESA: electrical signal spectrum analyzer; ADC: analog-to-digital converter; PCCB: program control circuit board; DAC: digital-to-analog converter; Driver: driver of the laser.

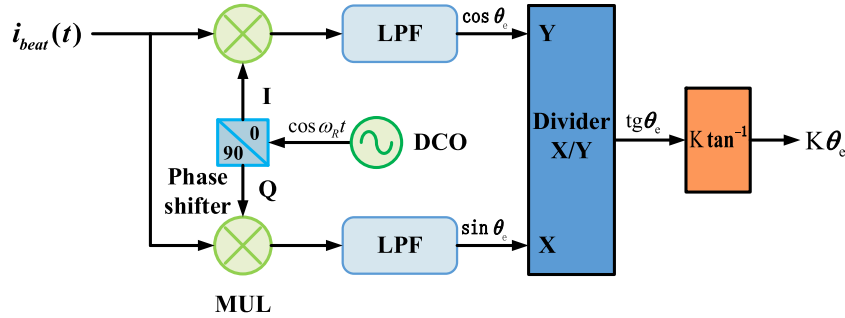


Fig. 2. Schematic diagram of the digital average discriminator. MUL: digital multiplier; DCO: digital control oscillator; LPF: low-pass filter (moving average filter).

3. Experiment of DFB-LD outputting frequency swept optical waves

In the experiments, we apply a commercial narrow linewidth DFB-LD of 1550 nm, with a linewidth of 175 kHz and a maximum output power of 60 mW. The delayed fiber length between the two arms of the MZI is 0.95 m. This delay is small enough to ensure that the phase errors caused by the laser nonlinearity can be corrected by the loop. The maximum locking range of the OPLL is ± 200 kHz. The maximum frequency swept bandwidth of this OPLL-based DFB laser is about 70 GHz. Experimentally, the linearity of the laser frequency swept signal is characterized by measuring the frequency spectrum or time-frequency curve of the beat signal, and the actual frequency swept curve of the laser can also be derived using the Hilbert transform of the beat signal. As shown in Fig. 1, we use an oscilloscope (OSC) (MSOS804A, Keysight Technologies) to acquire the beat frequency signal, and then calculate the time-frequency curve of the beat signal by the zero-crossing algorithm. The spectrum of the beat signal can be observed in real time by an electrical signal spectrum analyzer (ESA) (N9010, Agilent). The frequency swept period of the DFB-LD is 120 μ s, the bias current of the pre-distortion drive signal is 200 mA, and the amplitude is ± 85 mA, all of these parameters are implemented with PCCB control.

During the experiment, the triangular wave drive voltage applied to the DFB-LD driver is shown in Fig. 3(a), which is used as the initial drive voltage waveform without pre-distortion, and generates the time-frequency curve of the beat signal at the MZI output as shown in Fig. 3(d), from which it can be seen that the time-frequency curve is not a constant frequency value within a modulation period, but varies slowly with time over a wide bandwidth. The maximum frequency

change of 2.71 MHz in a modulation period indicates that the frequency of the DFB-LD output optical wave does not vary linearly with time. This conclusion is also verified later in Fig. 5(a) and (d). Fig. 3(b) shows the drive voltage waveform after 4th-order iterative pre-distortion and inflection point correction, where the 4th order pre-distortion waveform is calculated by the iterative pre-distortion algorithm [23], and we derive a general formula for iterative pre-distortion by establishing a relationship between the laser output frequency and the drive current. This iterative formula can derive a new pre-distortion drive current based on the drive current obtained from the last pre-distortion and the time-frequency curve of the beat signal. Similarly, the four-order iterative pre-distortion can be completed. For the inflection point correction, we achieve it by artificially replacing the waveform shape at the inflection point in the drive current waveform. From Fig. 3(e), it can be seen that there is a low-frequency component at the inflection point of the driven waveform, which is formed due to the triangular or triangular-like frequency sweep. Without considering the low-frequency component, it can be seen that the frequency range of the time-frequency curve of the beat signal is reduced to 313 kHz, indicating that the iterative pre-distortion algorithm is very effective in correcting large nonlinear tunable lasers. For the residual 313 kHz frequency fluctuation, which is caused by the residual frequency swept nonlinearity and random frequency noise in the system after iterative pre-distortion, it can only be suppressed by closed-loop phase locked. Fig. 3(c) shows the drive voltage waveform after closed-loop phase locked. From Fig. 3(f), it can be seen that the frequency variation range of the time-frequency curve of the beat signal further decreases and tends to a fixed value, which indicates that the DFB-LD output frequency swept optical waveform has a high linearity.

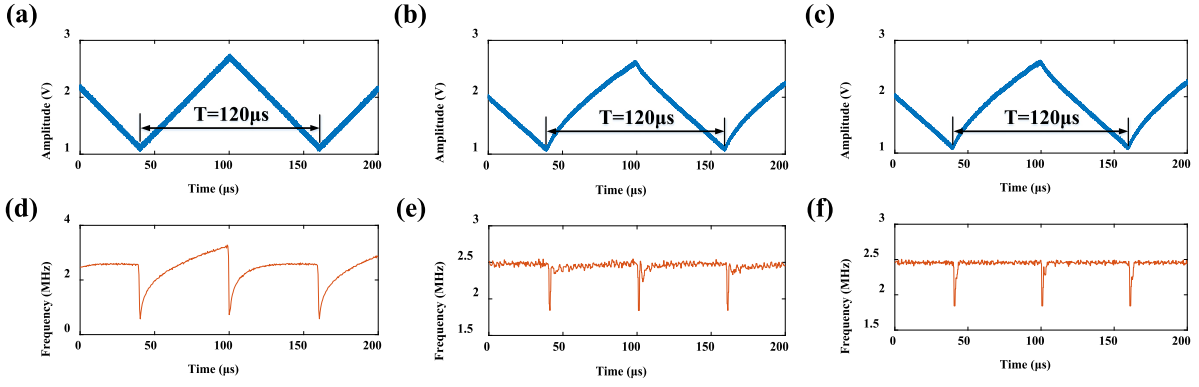


Fig. 3. (a)–(c) show the triangular wave drive voltage, the pre-distortion drive voltage, and the drive voltage after phase locking, respectively. (d)–(f) are the time–frequency curves of the beat signals corresponding to (a)–(c), respectively.

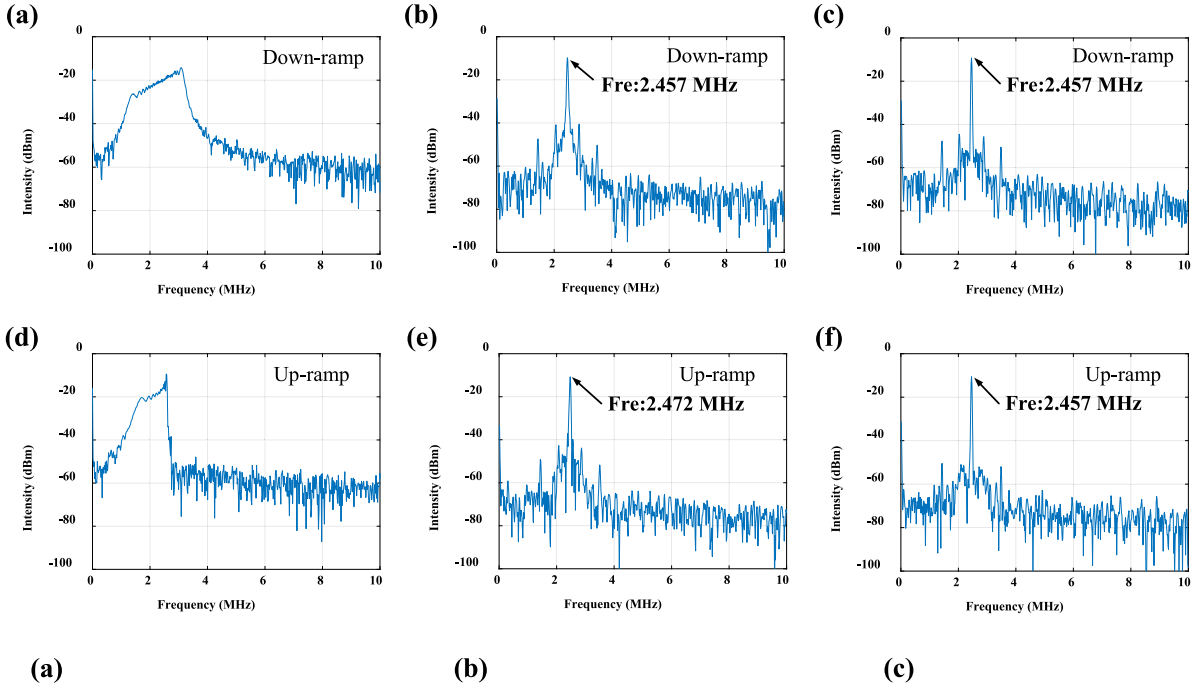


Fig. 4. The spectrum of the beat signal. (a) and (d) show the spectrum of the triangular-wave voltage-driven down-ramp and up-ramp corresponding beat signals. (b) and (e) are the spectra of the 4th-order pre-distortion voltage-driven down-ramp and up-ramp corresponding beat signals, respectively. (c) and (f) are the spectra of the down-ramp and up-ramp signals after closed-loop phase locking.

The spectrum of the beat signal under three voltage driving states was analyzed, where the down-ramp and up-ramp indicate the down-sweep and up-sweep of the DFB-LD output optical wave frequency. Fig. 4(a) and (d) gives the spectrum of the beat signal corresponding to the down-ramp and up-ramp of the DFB-LD under the triangular wave voltage drive. From the figures, it can be seen that there is a large nonlinearity for DFB-LD frequency swept under triangular wave voltage drive, which leads to a large expansion of the beat signal spectrum, and this state cannot be applied to the ranging system. When iterative pre-distortion is applied to the drive waveform, the beat signal spectra of laser frequency down-sweep and up-sweep are shown in Fig. 4(b) and (e), which are very much improved relative to the spectra in Fig. 4(a) and (d), and the frequency peaks can be found very easily. Because of the difference in the nonlinear characteristics of the DFB-LD during down-ramp and up-ramp, there is also a difference in the pre-distortion correction performed by the DFB-LD. The difference is manifested in the frequency spectrum as the peak frequencies of the down-ramp and up-ramp are not the same. When the system is closed-loop phase-locked, the signal-to-noise ratio of the down-ramp and up-ramp beat signals

Table 1
Stability experimental results.

		Group I	Group II	Group III
Down-ramp	1-r ²	3.65×10^{-8}	3.41×10^{-8}	3.53×10^{-8}
	RMS	1.39 MHz	1.35 MHz	1.37 MHz
Up-ramp	1-r ²	2.52×10^{-8}	2.23×10^{-8}	2.16×10^{-8}
	RMS	1.16 MHz	1.10 MHz	1.08 MHz

has a 10 dB improvement over Fig. 4(b) and (e), and the signal-to-noise ratio of the beat signals reaches 43 dB, and the same peak frequency of the down-ramp and up-ramp beat signals is achieved, as in Fig. 4(c) and (f).

When the DFB-LD outputs a triangular frequency swept optical wave, the time–frequency curve of the beat signal will have an inherent low-frequency component at the inflection point portion, as shown in Fig. 3(f). Therefore, when calculating the linearity of the frequency sweep of the DFB-LD in the three driving states, only the 80% region of interest (ROI) in the middle of the down-ramp and

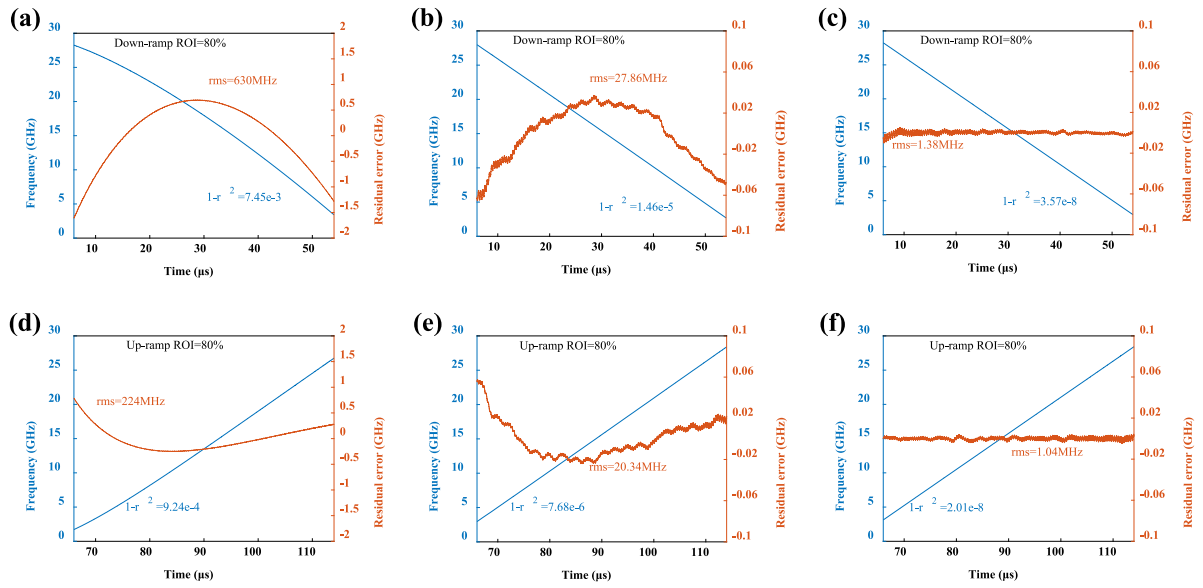


Fig. 5. Experimental results of DFB-LD linear frequency swept. Fig. (a) and (d) show the swept waveforms and the corresponding residuals in the down-ramp and up-ramp ROI of the laser output optical waveform driven by the triangular wave voltage. Fig. (b) and (e) show the frequency swept waveforms and the corresponding residuals in the down-ramp and up-ramp ROI of the laser output optical waveform driven by the pre-distortion voltage. Fig. (c) and (f) show the frequency swept waveforms in the down-ramp and up-ramp ROI of the laser output optical waveform and the corresponding residuals after closed-loop phase locking. The blue curve in the figure indicates the laser frequency swept waveform, which corresponds to the left vertical axis, and the brown curve indicates the residual error, which corresponds to the right vertical axis.

Table 2

Performance comparison of different linear frequency swept methods.

Ref.	Group/Year	Specifications of frequency-swept laser sources			Results of the calibration	
		Frequency swept excursion	Frequency swept slope	Repetition rate	1-R ²	RMS
[15]	Satyan's group 2009	100 GHz	100 THz/s	1 kHz	No	No
[17]	Wu's group 2021	26 GHz	52 THz/s	1 kHz	5.19×10^{-8}	1.5 MHz
[18]	Wu's group 2019	45 GHz	360 THz/s	4 kHz	1.8×10^{-8}	1.4 MHz
[19]	Satyan's group 2012	400 GHz	200 THz/s	250 Hz	No	No
[20]	Roos's group 2009	4.8 THz	6 THz/s	1.25 Hz	No	No

up-ramp is calculated, while the data generated near the inflection point are automatically sieved out. The detailed frequency swept curves of the down-ramp and up-ramp ROI of the DFB-LD output optical wave and their residuals are given in Fig. 5. From Fig. 5(a) and (d), under the initial triangular wave voltage drive, the down-ramp and up-ramp linearities $1 - r^2$ are 7.45×10^{-3} and 9.24×10^{-4} , respectively, corresponding to the frequency swept nonlinear RMS of 630 MHz and 224 MHz, both in the order of several hundred MHz. After the 4th-order pre-distortion, the frequency swept nonlinearity has been improved considerably, as can be seen from Fig. 5(b) and (e), where the frequency swept linearity $1 - r^2$ is improved by two orders of magnitude and the frequency swept nonlinearity RMS is reduced one order of magnitude, in the order of tens of MHz. It is worth mentioning that when the system is closed-loop phase-locked, the frequency swept linearity is improved again, and as can be seen from Fig. 5(c) and (f), the frequency swept linearity $1 - r^2$ reaches the 10^{-8} level, which is at least 2 orders of magnitude higher than the 4th order pre-distortion, and the frequency swept nonlinear RMS reaches the lowest level of 1.04 MHz.

For this method we designed stability experiments, which lasted for six hours and collected three groups of experimental data, each group being separated by three hours. The results of the stability experiments of the method are given in Table 1. The results show that the laser can maintain a high linearity of the frequency swept output throughout the six hours of the phase-locked experiment.

Finally, we present a comparison of our method with other linear frequency swept methods in Table 2. The results show that the use of digital phase locked method can achieve high frequency swept linearity at high frequency swept slopes, and the method allows real-time feedback as well as autonomous adjustment during the experiment to avoid external interference.

4. Conclusions

In this paper, we propose a digital optical phase-locked loop that implements linearization of laser frequency swept. The phase-locked loop has no specific requirements for lasers, and its linearization can be achieved even for lasers with large frequency swept nonlinearity. The effectiveness of the method was determined through repeated experiments with a 1550 nm DFB-LD. The method achieves a stable frequency swept output of the laser for a long time, and avoiding external interference. In the closed-loop phase-locked state, we achieve a frequency swept bandwidth of 31.1 GHz, and the swept slope is 518 THz/s, and a signal-to-noise ratio of 43 dB for the beat signal. In addition, we analyzed the linearity of the frequency swept after phase-locked, and the results showed that the maximum RMS value of the frequency swept nonlinearity is less than 1.4 MHz.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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